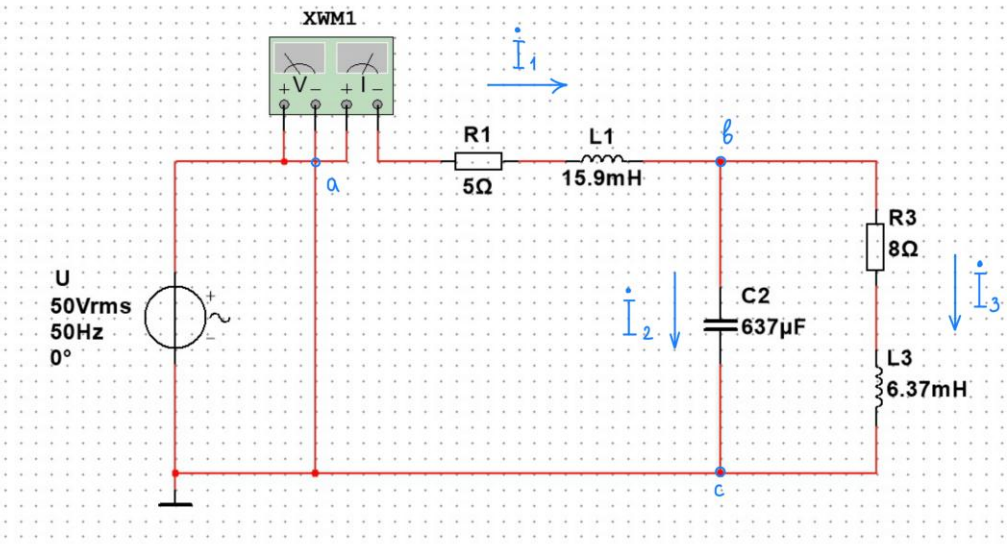


$\dot{U} = 50$ В

Найдем X_C и X_L для ветвей:

$$X_{L1} = 2\pi f L_1 = 2 \cdot 3{,}14 \cdot 50 \cdot 15{,}9 \cdot 10^{-3} = 4{,}995 \text{ Ом}$$

$$X_{C_2} = \frac{1}{2\pi f C_2} = \frac{1}{2 \cdot 3{,}14 \cdot 50 \cdot 637 \cdot 10^{-6}} = 4{,}997 \text{ Ом}$$

$$X_{L3} = 2\pi f L_3 = 2 \cdot 3{,}14 \cdot 50 \cdot 6{,}37 \cdot 10^{-3} = 2{,}001 \text{ Ом}$$

Найдем общие сопротивления ветвей:

$$\varphi_1 = \operatorname{arctg} \frac{4{,}995}{5} = 44^\circ 6'$$

$$\tilde{Z}_1 = R_1 + jX_{L1} = 5 + 4{,}995j = 7{,}1 e^{i 44^\circ 6'} \text{ Ом}$$

$$\varphi_2 = -90^\circ$$

$$\tilde{Z}_2 = -jX_{C_2} = -4{,}997j = 4{,}997 e^{-i 90^\circ} \text{ Ом}$$

$$\varphi_3 = \operatorname{arctg}\left(\frac{2{,}001}{8}\right) = 14^\circ 2'$$

$$\tilde{Z}_3 = R_3 + jX_{L3} = 8 + 2{,}001j = 8{,}247 e^{i 14{,}04^\circ} \text{ Ом}$$

Найдем эквивалентное сопротивление

$$\tilde{Z}_{23} = \frac{\tilde{Z}_2 \cdot \tilde{Z}_3}{\tilde{Z}_2 + \tilde{Z}_3} = \frac{-4{,}997j\,(8 + 2{,}001j)}{-4{,}997j + 8 + 2{,}001j} = 2{,}7374 - 3{,}9719j \text{ Ом}$$

$$\tilde{Z} = \tilde{Z}_1 + \tilde{Z}_{23} = 5 + 4{,}995j + 2{,}7374 - 3{,}9719j = 7{,}7374 + 1{,}02321j \text{ Ом}$$

$$|\tilde{Z}| = \sqrt{7{,}7374^2 + 1{,}02321^2} = 7{,}806\ \text{Ом} \quad \varphi = \operatorname{arctg}\left(\frac{1{,}02321}{7{,}7374}\right) = 7{,}57^\circ$$

$$\tilde{Z} = 7{,}806 e^{j7{,}57^\circ}\ \text{Ом}$$

$$\dot{I}_1 = \frac{\dot{U}}{\tilde{Z}} = \frac{50}{7{,}806 e^{j7{,}57^\circ}} = \boxed{6{,}405 \cdot e^{-j7{,}57^\circ}\ \text{A}}$$

$$\dot{U}_{bc} = \tilde{Z}_{23} \cdot \dot{I}_1 = (2{,}7374 - 3{,}9719j)\, 6{,}405 \cdot e^{-j7{,}57^\circ} =$$

$$4{,}824 e^{-j55{,}43^\circ} \cdot 6{,}405 \cdot e^{-j7{,}57^\circ} = 30{,}898 \cdot e^{-j \cdot 63^\circ}\ \text{B}$$

$$\dot{I}_2 = \frac{\dot{U}_{bc}}{\tilde{Z}_2} = \frac{30{,}898 \cdot e^{-j \cdot 63^\circ}}{4{,}997 e^{-j90^\circ}} = \boxed{6{,}183 \cdot e^{j \cdot 27^\circ}\ \text{A}}$$

$$\dot{I}_3 = \frac{\dot{U}_{bc}}{\tilde{Z}_3} = \frac{30{,}898 \cdot e^{-j \cdot 63^\circ}}{8{,}247 e^{j14{,}04^\circ}} = \boxed{3{,}747 e^{-j77{,}04}\ \text{A}}$$

Найдем мощности цепи и ветвей:

$$\tilde{S} = \dot{U} \cdot \dot{I}_1 = 50 \cdot 6{,}405 \cdot e^{+j7{,}57^\circ} = 320{,}25 \cdot e^{+j7{,}57^\circ}\ \text{BA} =$$

$$320{,}25\,(\cos(+7{,}57^\circ) + j\sin(+7{,}57^\circ)) = 317 + 42j\ \text{BA}$$

$P = 317$ Вт $\quad Q = +42$ ВАр

Найдем активную мощность в ветвях:

$$P_1 = R_1 \cdot I_1^2 = 5 \cdot 6{,}405^2 = 205{,}120\ \text{Вт}$$

$$P_2 = 0$$

$$P_3 = R_3 \cdot I_3^2 = 8 \cdot (3{,}747)^2 = 112{,}320\ \text{Вт}$$

Проверка:

$$P = P_1 + P_2 + P_3 = 205{,}120 + 112{,}320 = 317{,}44\ \text{Вт} \approx 317\ \text{Вт}$$

Найдем реактивную мощность в ветвях:

$$Q_1 = X_{L1} \cdot I_1^2 = 4{,}995 \cdot 6{,}405^2 = 204{,}915\ \text{ВАр}$$

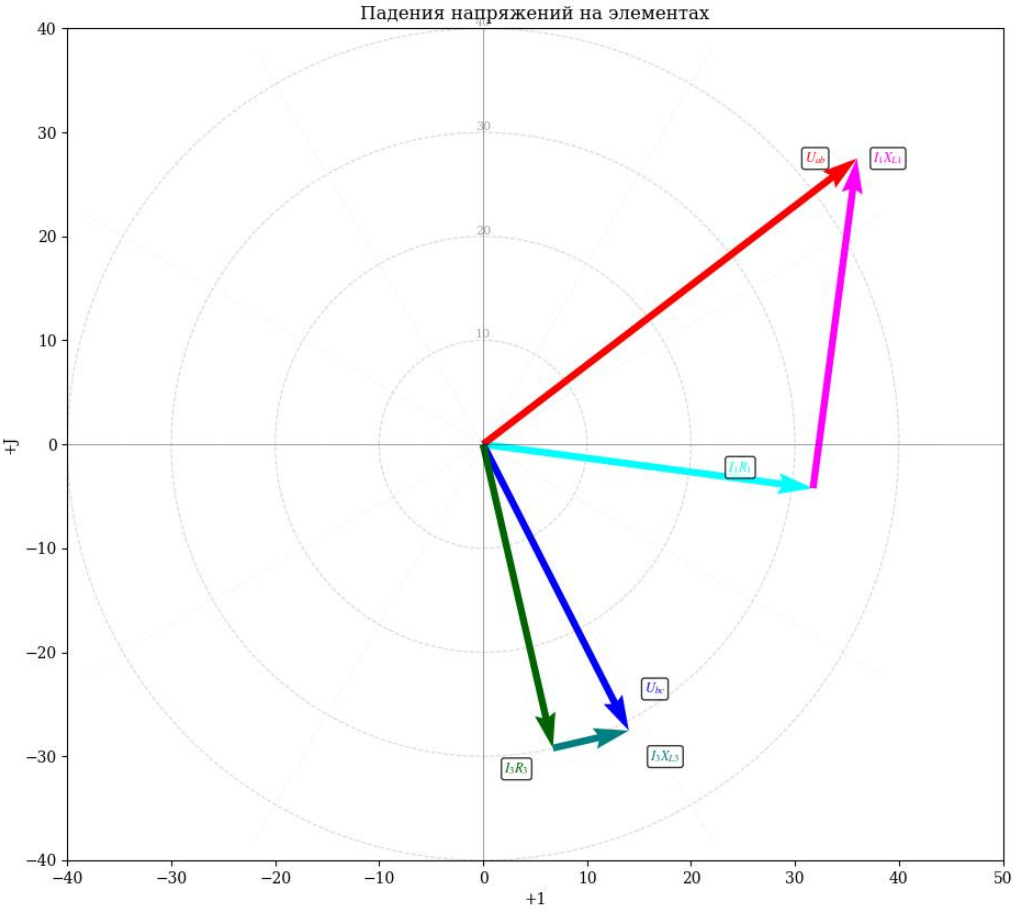
$$Q_2 = -X_{C2} \cdot I_2^2 = -4{,}997 \cdot 6{,}183^2 = -191{,}033\ \text{ВАр}$$

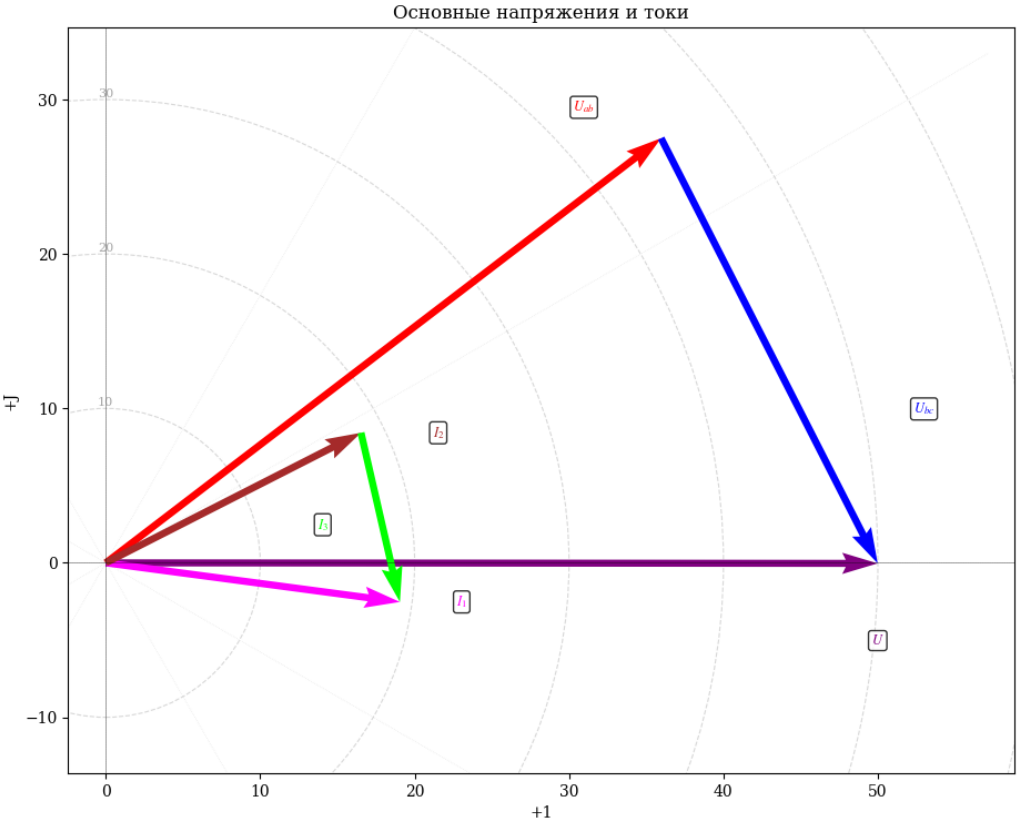
$$Q_3 = X_{L3} \cdot I_3^2 = 2{,}001 \cdot 3{,}747^2 = 28{,}094\ \text{ВАр}$$

Проверка:

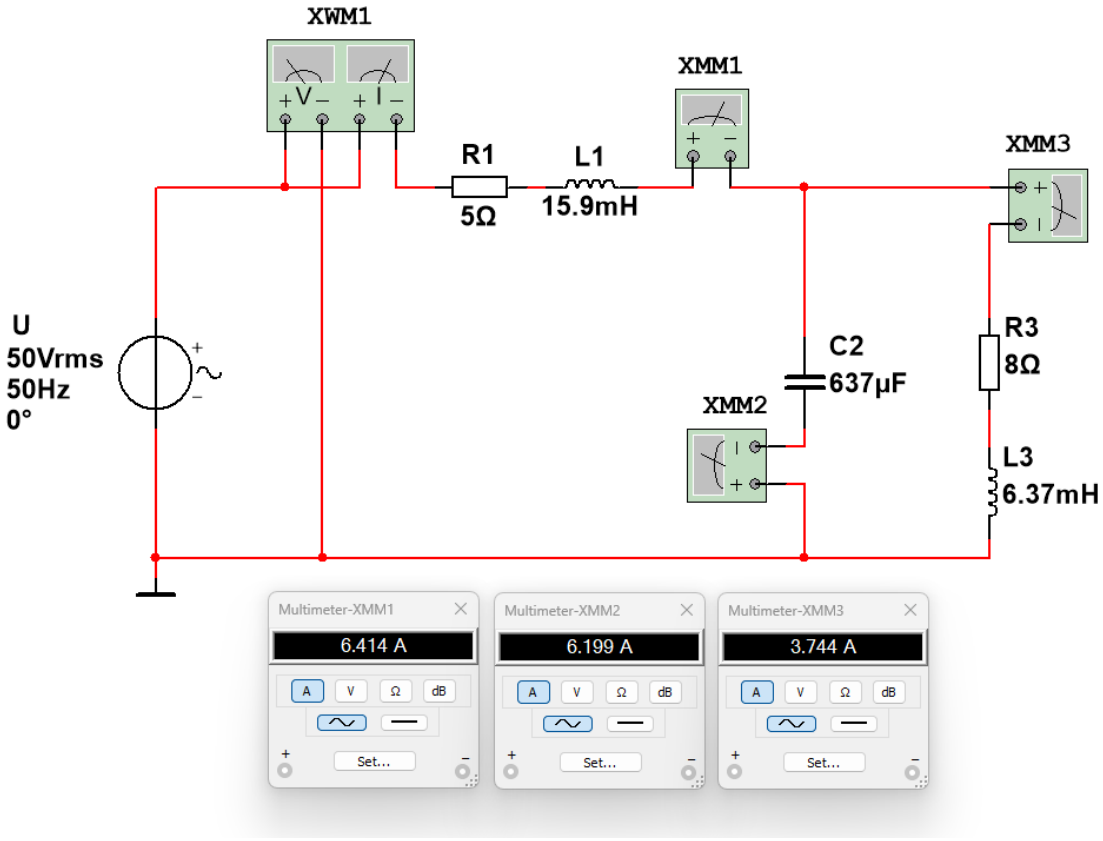
$$Q = Q_1 + Q_2 + Q_3 = 204{,}915 - 191{,}033 + 28{,}094 =$$

$$41{,}976\ \text{ВАр} \approx 42\ \text{ВАр}$$

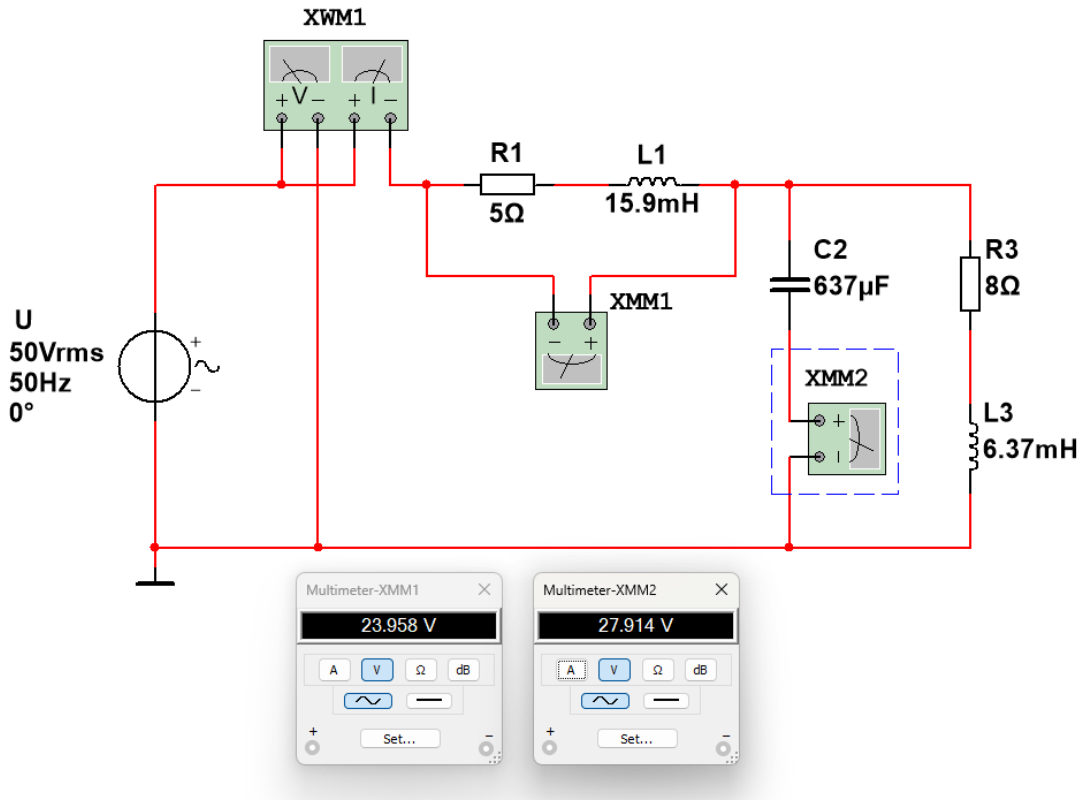

Падения напряжений на элементах
U_{ab}
I_1X_{L1}
I_1R_1
U_{bc}
I_3X_{L3}
I_3R_3
+J
+1


Основные напряжения и токи
U_{ab}
U_{bc}
I_2
I_3
I_1
U
+J
+1

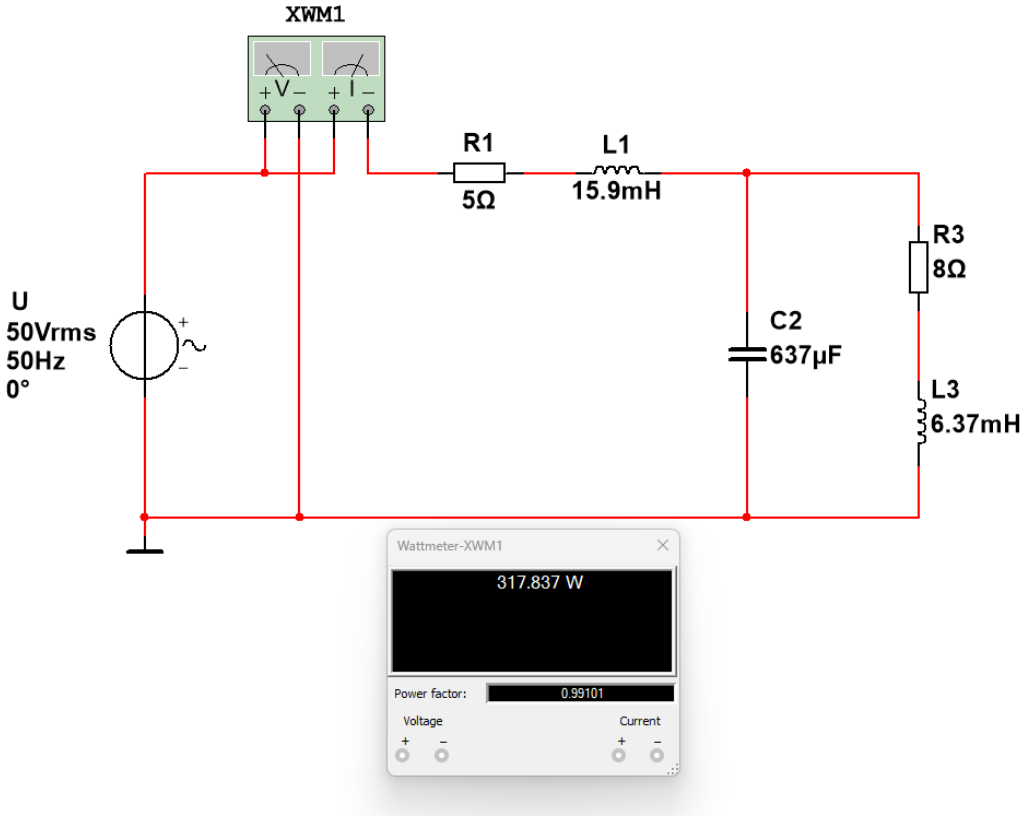
Измерим токи ветвей:



Измерим напряжения ветвей:



Измерим активную мощность:



Итого:

Величина	Рассчитано	Измерено
I_1	6.405 A	6.414 A
I_2	6.183 A	6.199 A
I_3	3.747 A	3.744 A
U_{bc}	30.898 В	27.914 В
P	317 Вт	317.837 Вт